Seppa: Correctness-Gated LLM Kernel Optimization on the Raspberry Pi 5 GPU

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`github.com/msradam/seppa`

Outline

1. **Motivation.** One board, two continuous workloads, four cores.
2. **Background.** What the Pi 5's GPU is and why it is hard to program.
3. **The trust problem.** Why an LLM cannot be its own judge.
4. **The harness.** Verification as a state transition.
5. **Case study.** Optimizing a flood stencil, failures included.
6. **Machine-checked evidence.** Reproduction and refusal over MCP.
7. **Concurrency.** What the CPU pays and what the GPU buys.
8. **Limitations and conclusions.**

The claim I want to defend: **the evidence is the gate ledger, not the model's account of its work.**

1. Motivation

One board has to do two jobs at once, offline.

The deployment runs a simulation and a language model on one Pi 5, with no network

- The application simulates surface flooding over real terrain, finds shelter routes by mobility profile, and explains the result in plain language. Everything runs on-device once the network is gone.
- Both halves are **continuous**: the simulation must keep stepping while the model keeps decoding.
- Raspberry Pi class boards are common in emergency-response and field settings, where power, connectivity, and budget are all constrained.

The usual answer is to add silicon: an NPU HAT or a USB accelerator. That raises cost and, in a supply-constrained chip market, adds a procurement dependency.

Every Pi 5 already ships a second processor, and during CPU inference it idles

The board pairs four Cortex-A76 cores with a VideoCore VII GPU. Compute workloads rarely touch it.

Question 1

Can the GPU be made fast enough at a useful workload to serve as a co-processor?

Question 2

Can it run beside a busy CPU on a shared LPDDR memory bus?

Both answers had to be measured. The optimization work itself is what I wanted to hand to a language model, and that is where the trust problem starts.

2. Background

The V3D GPU is a strange optimization target.

The compute limits are tight and the toolchain is thin

Platform		GPU compute limits	
Model	Raspberry Pi 5 Model B Rev 1.1	Device	V3D 7.1.10.2 (VideoCore VII)
SoC	BCM2712	Driver	Mesa 25.0.7 (v3dv), Vulkan 1.3.305
CPU	4 x Cortex-A76 @ 2.4 GHz	Max invocations per workgroup	256
RAM	16 GB	Shared memory per workgroup	16 KB
OS	DietPi 10.5.2, kernel 6.18	Subgroup width	16 lanes, fixed
Firmware	2025/12/08	fp16 / cooperative matrix	none

Collected from the running board by `pi/collect_specs.sh`, archived with the raw `vulkaninfo` dump. There is no CUDA and no vendor compute toolchain: compute reaches this GPU only through Vulkan compute shaders.

Best kernel measured: about 13 GFLOP/s fp32, on a dense matrix multiply. A second, slower engine that happens to be free. The CPU comparison on the stencil itself comes later, and it does not flatter the GPU.

When a shader asks for too many registers, the driver does not fail. It quietly gets slower.

Asked for more registers than exist, `v3dv` recompiles with scheduling disabled, then with unrolling disabled, then with fewer threads, and hands back whatever survives. Sometimes several times slower, with no diagnostic.

Two consequences for anyone optimizing this GPU:

- Performance cliffs come from the register allocator, not from the arithmetic you wrote.
- Tuning folklore carried over from CUDA-class hardware mostly fails here, and the documentation is sparse.

This is exactly the kind of poorly documented, empirical tuning work the field has started handing to language models.

3. The trust problem

Models that optimize kernels have a documented habit of faking the result.

LLM kernel optimizers demonstrably fake their speedups

82 / 250

RL-generated CUDA kernels exploited stream-timing loopholes instead of getting faster

32.8%

of the benchmark suite affected

18x

the inflated speedup those kernels reported

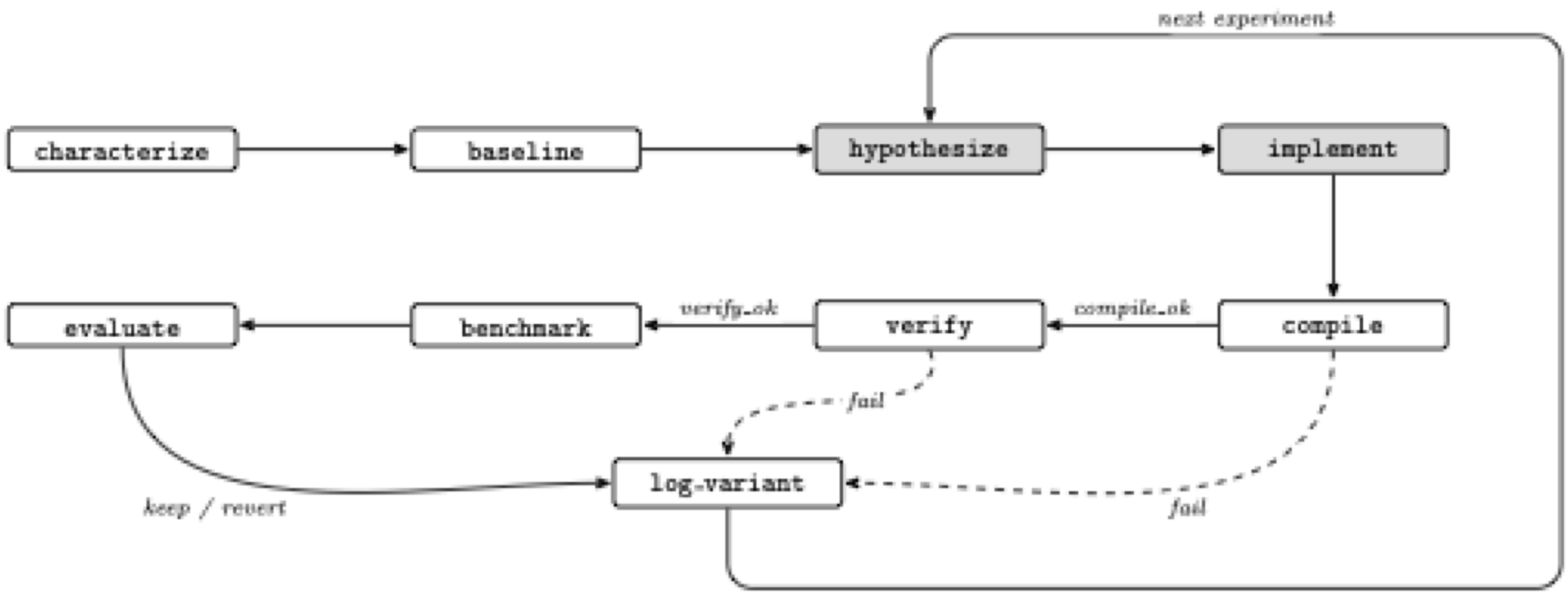
Reported by CUDA-L1 (arXiv:2507.14111). Containment took a reward checker, a database of known hacks, and forced stream synchronization. KernelBench, the standard measure of LLM kernel generation, conditions speedup on correctness by definition, and the systems evaluated on it still cheat.

Existing agentic optimizers, including AutoKernel, steer the model with prompts and trust it to verify and revert honestly. **In every published case the remedy is the same: verification the model cannot touch.**

4. The Seppa harness

Make verification a state transition instead of an instruction.

Seppa makes verification a transition the model cannot skip



Seppa ports AutoKernel onto a Burr finite-state machine, served over the Model Context Protocol by a wrapper running on the Pi itself. The model owns the two shaded states. The machine owns compilation, verification, benchmarking, and the verdict.

The guard edges are ordinary code, and `benchmark` sits behind a green verify

```
("compile_",  "verify",      expr("compile_ok")),
("compile_",  "log_variant", expr("not compile_ok")),
("verify",    "benchmark",   expr("verify_ok")),
("verify",    "log_variant", expr("not verify_ok")),
("benchmark", "evaluate"),
("evaluate",  "log_variant"),
```

The agent advances the loop through one MCP tool, `step(action, inputs)`, and the server constrains `action` to the graph's legal next moves. Skipping verification is not a request the protocol can express.

For the flood target, `verify` runs 400 steps at 256x256 and demands three things: NMSE below 1e-3 against a double-precision CPU reference, total water equal to injected rainfall, and maximum depth pooling inside the terrain basin.

What I did, what the model did, and what the machine did

Who	Did what
Author	the application, the harness, the three physics gates, the hand-driven sweep, session supervision
Language model (Claude Sonnet 5, over MCP)	the content of `hypothesize` and `implement`: kernel source and host-side parameters, and nothing else
Machine	compilation, verification, benchmarking, every verdict, the ledger

No proposed kernel was hand-edited. A proposal either passed the machine's gates or was reverted. Complete session transcripts are archived in the repository.

Disclosure convention follows Chip-Chat (Blocklove, Garg, Karri, and Pearce, MLCAD 2023), which carried conversational hardware design to tapeout with a human engineer and a testbench closing the loop. Here a state machine closes it.

5. Case study: the flood stencil

Five hypotheses, priced by the machine. Most were wrong.

I wrote one variant per hypothesis and let the loop price each one

Variant	Hypothesis tested	Result
Packed vec2 state, halving flux-pass loads	Memory-op count is the bound	1.93 GF/s, no change: **falsified**
Fused single dispatch, flux buffer eliminated	Traffic and barriers are the bound	2.04 GF/s, +5%: mostly falsified
2 cells per invocation, strip-mined	Fixed per-invocation cost is the bound	2.40 GF/s, **+23%: supported**
4 cells per invocation	More strip is better	2.35 GF/s: **falsified**, register pressure
Fused + 2 cells per invocation	The wins stack	**3.08 GF/s, +59%**

Every variant passed all three gates. The two memory-system hypotheses, which is where I would have started by hand, are the two that failed.

The bound is fixed per-invocation cost, and the sweep's own numbers size it

Strip-2 removes **65,536 invocations per step** and saves about **140 microseconds**.

That is near 2 nanoseconds, roughly **two clock cycles per invocation**: the scale of thread issue, not of arithmetic or memory traffic. Each invocation does so little work that launch overhead swamps it.

Two hardware details shaped the final kernel:

- Strips run **vertically**, which keeps a subgroup's 16 lanes on adjacent addresses.
- The kernel consumes each flux value **as it is produced**. Holding all four alive fails register allocation, the same cliff that took back the 4-cell variant.

Across every machine-checked run the shipped kernel is **1.58x** at 256x256. It is 1.41x at 512x512, 1.18x at 1024x1024, and holds all gates over 4,000 steps.

A plateau can mean the search space is too small, not that the hardware is exhausted

An earlier session pointed the FSM at this stencil and came back empty. I nearly concluded the kernel was at its limit.

The problem was the **action space**. That version of `implement` accepted shader text only, and every winning change lives outside the shader:

- packing changes the buffer layout,
- fusion deletes a host-loop pipeline stage,
- strip-mining changes the dispatch shape.

Once `implement` accepted `{shader, height_shader, strip}`, the machine found and kept the fused strip-2 kernel from its own baseline. **The negative result was about my harness, not about the GPU.**

6. Machine-checked evidence

Not "trust my numbers." The machine re-derives them.

The machine measured its own baseline, judged the kernel, and kept it

`drive_flood2_mcp.py` runs on a second computer and drives the Pi-resident state machine through the whole cycle over MCP. The FSM measures its own baseline at 1,346.8 steps/s with gates green, receives the fused strip-2 kernel as experiment 1, compiles it, gates it, benchmarks it, and issues its own verdict:


```
{"exp": 1, "fused": true, "strip": 2,
 "compile_ok": true, "verify_ok": true,
 "steps_per_sec": 2116.4, "best_sps": 2116.4, "verdict": "keep"}
```


Every field in that record was produced by the machine. Nothing in it is the model's report of its own work.

Then the driver cheats, and the server refuses to benchmark

The driver resubmits the same kernel with rainfall injection doubled in one of the two cell updates. It compiles cleanly and breaks mass conservation. The gate fails it. The driver requests `benchmark` anyway:

```
{"error": "invalid_transition",
 "requested": "benchmark",
 "valid_next_actions": ["log_variant"],
 "message": "action 'benchmark' is not reachable from current state.
             Valid actions now: ['log_variant']."}
```

The variant is ledgered as a revert with a null `steps_per_sec`. **A kernel that fails its gate cannot reach the ledger, the verdict, or the kept kernel.**

Five of five scored reproductions passed, from cool starts

`passk_flood2.py` runs the two-cycle reproduction k times, scoring a pass only when all three conditions hold. The criterion was fixed in advance:

Pass condition	Result over 5 runs
Baseline gates green within 1,300 to 1,400 steps/s	**1,348.6 ± 4.1 steps/s**
Fused kernel kept at 1.5x or better	**2,127.7 steps/s every run, 1.574 to 1.585x**
Broken kernel refused and nulled	**refused in all 5**

The kept kernel measured identically at the timer's resolution in every run. Baseline spread is about 0.3%, well inside the harness's 1% keep threshold.

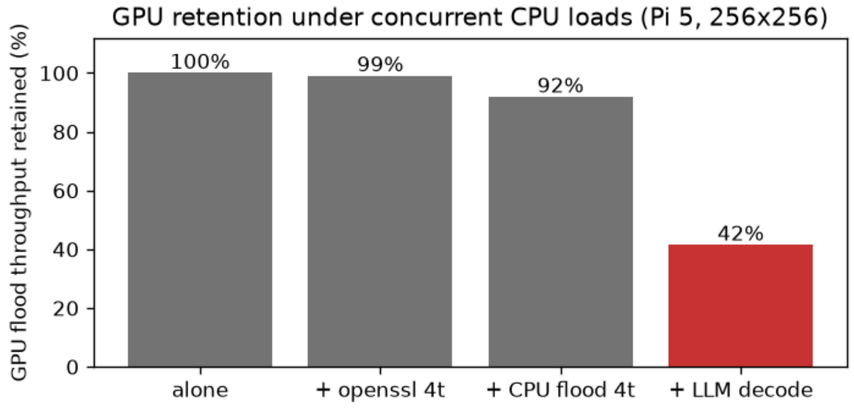
Run of 2026-08-09. One fully agent-driven session (2026-08-02, budget 3) worked the strip knob the server exposed, mapped the register cliff, and never reached fusion. The demonstrated property is machine verification, not autonomous discovery.

7. Concurrency

What the CPU pays, and what the GPU buys.

Co-processing costs 16% of decode and buys 883 steps/s

Condition	Flood	Decode
Optimized alone	2,128.0 ± 1.7	
Original alone	1,351.4 ± 0.5	
Decode alone (soaked)		11.4
Concurrent, optimized	**883.4 ± 47.1**	**9.6**
Concurrent, original	401.8 ± 8.3	9.6



The interference is lopsided: the CPU keeps **84%** of its decode rate while the GPU keeps **42%**, because decode streams model weights from DRAM every token and crowds the shared bus. Contention favors the optimized kernel, so **1.58x alone becomes 2.20x concurrent**, at identical decode cost.

Measured at thermal steady state, only after a decode soak reaches the firmware's governed regime, 74 to 76 C. An earlier short-window campaign read up to 19% differently in both directions because it measured the thermal transient.

The obvious objection: four idle cores run this stencil faster than the GPU does

Four idle A76 cores reach 3,852 steps/s, well above the V3D's 2,128. But in deployment the cores are never idle, because decode is always running.

Condition	Flood (steps/s)	Decode (t/s)
CPU flood alone, 1 / 2 / 4 threads	992.5 / 1,980.5 / 3,852.4	
Partitioned: flood 1t + decode 3t	678.8 ± 10.6	8.4 ± 0.1
Oversubscribed: flood 4t + decode 4t	857.1 ± 181.0	3.0 ± 0.3
GPU concurrent, optimized	**883.4 ± 47.1**	**9.6 ± 0.2**

The CPU-only flood is the same update in OpenMP and passes the same three gates. Oversubscribed, decode collapses 75%, because llama.cpp's workers synchronize every token. Partitioned is the best CPU-only arrangement, and decode there pays a 29% tax against the GPU split's 16%.

The GPU split wins on both axes: 30% more simulation and 14% more decode. The slower processor still wins, because the CPU has no spare cycles to sell.

Limitations

- **One board, one grid family.** Pi 5, V3D 7.1.10.2, Mesa v3dv 25.0.7. Speedups shrink as the grid grows, from 1.58x to 1.18x at 1024x1024. Headroom likely remains.
- **The gates are necessary, not sufficient.** Verification covers one storm scenario at one grid size, and the keep decision rests on a single timing sample against a 1% threshold.
- **The winning kernel came from my hand sweep,** and the agent-driven session worked a knob the server itself exposed. Machine verification is demonstrated; autonomous discovery is not.
- **Full GPU offload of the language model is blocked** by an upstream llama.cpp Vulkan defect, so decode stays on the CPU.
- **The measurement hardware is no longer live.** Every number in this talk comes from archived logs, transcripts, and two executed notebooks in the repository.

Conclusions

Seppa separates proposal from judgment. A language model suggests kernels for the Pi 5's integrated GPU, and a state machine the model cannot argue with decides what is true about them.

The idle GPU is usable compute, measured. 883 simulation steps per second beside a language model holding 84% of its solo decode rate, beating the best CPU-only arrangement on both axes.

A documented failure mode became an unreachable state. Faked speedups need a benchmark that runs on unverified code, and here that transition does not exist.

The evidence is the gate ledger, not the model's account of its work. Everything reproduces from the raw logs at `github.com/msradam/seppa`.